

## 4. Trade-offs: The Composite Pattern

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Implementing the Composite Pattern involves balancing specific pros and cons:

### Advantages

- **Uniformity:** Allows the system to treat single devices and groups of devices (rooms/floors) the same way.
- **Management:** Simplifies the management of complex hierarchies.
- **Flexibility:** Makes it easy to add new devices or entire rooms without disrupting existing code.

### Disadvantages

- **Complexity:** Large systems can become more complex to navigate.
- **Design Effort:** Requires careful initial structure design to ensure it functions correctly.